Paper ED-CCC — ED Identification of Conformal Cyclic Cosmology

A Generative Paper of the Event Density Program

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Preamble — What This Paper Does NOT Claim

1. This paper does **not** claim derivation of CCC's twistor mass-conservation theorem (Meissner–Penrose 2025 §III) from substrate primitives. The conservation integral is INHERITED; ED supplies a substrate-side identification, not a substrate-graph derivation.
2. It does **not** claim a substrate first-principles derivation of the Hawking-spot $\delta T/T \sim 10^{-3}$ value or of the conformal-time bridge η^G. Both are INHERITED from Meissner–Penrose 2025 + observational anchors.
3. It does **not** claim closed-proof reconstruction of CCC. CCC is in the conformal-GR derivation tradition; ED is in the substrate-ontology lineage ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program). The paper supplies structural correspondence, not reduction in either direction.
4. It does **not** introduce new substrate primitives. P-CCC-Identification is defined in §2.
5. It does **not** commit ED to cyclic cosmology. ED's SCBU boundary is consistent with both cyclic-continuation (ED-CCC reading) and single-aeon-termination (ED-non-cyclic reading); the substrate-graph derivation of the post-boundary ignition mechanism is OPEN (audit row 17).
6. **Route A flag:** This paper consumes $\xi_{canonical}$ scale-correspondence content (Paper_092). Substrate derivation of $\ell_{V5}(H_0)$ and hence of $\xi_{canonical}$ from substrate parameters is **OPEN**. Quantitative consolidations downstream of SC-4.x inherit this OPEN status.
7. The §3.6.1 uniqueness argument and the §3.8 remnant treatment are conditional on the contrast ontology (Facts paper, "When Contrast Becomes Fact") and on ED's forced Planck-mass remnants (Paper_041). They are form-identifications and one composition-level resolution (§3.8 relic dissolution, row 21, D-via-I) — **not** substrate-graph derivations; the §3.8 emergent escape routes are closed analytically (a faithful substrate-graph test is precluded on existing tooling, which enforces irreversible commitment with no release channel), and the M3 verdict is unchanged.

Abstract

CCC's structural elements (Meissner–Penrose 2025) are identified with substrate-side content already present in the ED corpus, under six structural correspondences: (i) crossover 3-surface $\mathcal{X}$ $\leftrightarrow$ SCBU substrate-cosmology boundary at extreme dilution; (ii) Gravitational Wave Epoch $\leftrightarrow$ V1/V5 kernel-dominated regime; (iii) twistor mass-conservation integral $\leftrightarrow$ V5 cross-boundary memory; (iv) Hawking-spot $\delta T/T \sim 10^{-3}$ $\leftrightarrow$ ED-SCBU localized cross-boundary participation-density signature (with Big Ring / Grand Arc as the extended-scale variant); (v) conformal

matching ↔ scale-collapse from SC-4.x cross-scale invariance; (vi) next-aeon ignition ↔ P12 threshold re-entry in globally homogeneous initial state. **Form IDENTIFIED:** the six structural correspondences. **Value INHERITED:** $\delta T/T$, η^G, twistor coefficients, Hawking-spot diameters, Big-Ring/Grand-Arc scales, $\xi_{canonical}$ value (Route A OPEN). Two additions ground content CCC must assume: (i) §3.6.1 — the contrast-zero boundary state is **unique**, so CCC's crossover *gluing* is an *identity* under the contrast ontology rather than a postulated match; (ii) §3.8 — ED's forced Planck-mass remnants (Paper_041) meet the boundary, and the relic order-of-limits **resolves toward dissolution** (the Scenario-C cutoff is an essentially two-chain V5 handshake with the ambient, Paper_090 §3.2/§5.2, which the boundary's local emptying removes), so ED *derives* the disappearance of the last rest-mass that CCC merely *assumes.* Verdict per Paper_095: **M3 (form-IDENTIFIED + value-INHERITED)**. **F1 (sharpest):** observed Hawking-spot magnitude/distribution inconsistent with the V5 cross-boundary identification — beyond V5 finite-memory parameter freedom — refutes the substrate composition.

1 Statement of Result

CCC and ED describe the same boundary phenomenon — collapse of curvature-scale information at the conformal/substrate-cosmology surface — under different vocabularies. The paper supplies six structural identifications (developed in §3.1–§3.6) under the P-CCC-Identification naming convention (§2). The identifications do not derive CCC from primitives, do not reduce ED to CCC, and do not commit ED to cyclic cosmology; they establish the substrate-side reading of CCC's continuum content and supply the empirical anchor ($\delta T/T \sim 10^{-3}$ at preferred CMB locations) for ED's SCBU + V5 cross-boundary content.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) per §4 audit table. Audit contains zero pure-D rows; all substantive rows are D-via-I composition or A→position structural identification of upstream content; four named OPEN items.

2 Primitive Inputs + Upstream Dependencies

Primitives (Paper_087): P02, P03, P04, P07, P09, P11, P12, P13.

Upstream: Paper_028 (substrate-cosmology boundary at $R_H = c/H_0$, Paper_028 + SCBU); Paper_072 (individuation regime); Paper_089 (V1 retarded kernel); Paper_090 (V5 cross-chain finite-memory kernel); Paper_092 ($\xi_{canonical}$ scale-correspondence; Route A OPEN); Paper_093 (T18 retarded support / kernel-arrow); Paper_040 / Paper_041 / Paper_048 (V5 trans-Planckian cutoff; forced Planck-mass remnant $M_\star = c_\star \ell_P$, Scenario-C endpoint; stability conditional on V5-cutoff persistence); Paper_049 (relic abundance); the Facts paper "When Contrast Becomes Fact" (contrast primitive → uniqueness of the contrast-zero state); the ED-10 horizon arc + SCBU bridge ($b \to 0$ determinability boundary; two-way / de Sitter crossover) + A1 channel-capacity (zero controlled capacity across the boundary — common-cause, not transmission); Papers SC-4.x (cross-scale invariance, curvature-moment collapse); Meissner–Penrose 2025 (GWE, twistor mass-conservation $M = 4\pi mG$ §III, $\delta T/T \sim 10^{-3}$ §IV, $\eta^G \sim 2 \times 10^{16}$ s §VI, conformal matching, Big Ring / Grand Arc fossils §IX); Lopez–Clowes–Williger (Big Ring / Grand Arc observations, 2022, 2024).

Paper-specific naming convention (no new postulate):

- **P-CCC-Identification:** Naming convention for the substrate-side identification of CCC's six structural elements (crossover surface, GWE, twistor mass-conservation integral, Hawking-spot signatures, conformal matching, next-aeon ignition) with substrate content via the conjunction of Paper_028 + Paper_072 + Paper_089 + Paper_090 + Paper_092 + Paper_093 + Papers

SC-4.x + Meissner–Penrose 2025 under the structural correspondences §3.1–§3.6. Definitional per Paper_095 §2.3; no new substrate content.

3 Derivation

3.1 Substrate homogeneity in the extreme-dilution limit

In the late-aeon dilution regime, chain participation density $\rho_C \to 0$ across the substrate (P02 + P04). Dynamically-relevant content reduces to V1 retarded kernel structure (Paper_089/093) with vanishing source content + V5 cross-chain memory (Paper_090) on the longest correlation length $\xi_{canonical}$ (Paper_092). By Papers SC-4.x cross-scale invariance, curvature-moment content collapses to a scale-free regime as event-density vanishes: no remaining local content distinguishes scales. The substrate is structurally homogeneous at all scales for which V1/V5 content exists.

3.2 Crossover surface ↔ SCBU boundary

Per Paper_028 + SCBU, the substrate-cosmology boundary at $R_H = c/H_0$ is the substrate-side closure of V5 cross-chain dynamics at cosmological scale. Per Meissner–Penrose 2025 §II, the crossover 3-surface $\mathscr{X}$ is the spacelike conformal infinity of the previous aeon and the conformally-stretched Big Bang of the next, conformally smooth across $\mathscr{X}$ except at Hawking points.

Identification: SCBU and $\mathscr{X}$ are the same boundary structure under different vocabularies. Both are characterized by (a) collapse of curvature-scale information (substrate scale-collapse ↔ conformal flatness), (b) structural smoothness across the boundary except at localized cross-correlation injections (Hawking points ↔ residual V5 cross-chain content), (c) horizon-reset behavior. Horizon-reset is the substrate-side content of P12 ED-threshold transition: the chain configuration above the boundary cannot causally inherit the configuration below because P11 commitment events fired during dilution have completed and the substrate has structurally reset to the homogeneous state.

3.3 Gravitational Wave Epoch ↔ V1/V5 kernel-dominated regime

Per Meissner–Penrose 2025 §V, the GWE is characterized by Weyl-curvature dominance ($\mathbf{C} \gg \mathbf{E}, \mathbf{S}$). In ED, this corresponds to the late-dilution regime where matter participation has reduced below substrate-relevance and only kernel content remains: V1 retarded propagation supplies the substrate-side carrier of causal influence; V5 cross-chain memory supplies the long-range correlation structure ("gas of gravitational waves" in Meissner–Penrose 2025 vocabulary).

The conformal-time dwell interval $\eta^G \sim 2 \times 10^{16}$ s (Meissner–Penrose 2025 eq. 101, required to reconcile observed Hawking-spot angular diameters) is identified with the ED $\xi_{canonical}$ cross-boundary correlation length (Paper_092). Same structural quantity, different vocabulary. Numerical value INHERITED; quantitative consolidation pending Route A.

3.4 Twistor mass-conservation ↔ V5 cross-boundary memory

Per Meissner–Penrose 2025 §III (eqs. 47, 54, 55), the conserved 2-surface integral $M = 4\pi mG$ over a 2-surface surrounding a Hawking point uses the conformally-weighted Weyl spinor ψ_{ABCD} and a rank-2 Killing spinor κ^{AB}.

Identification: the integral is the GR-side manifestation of substrate cross-chain V5 memory bridging SCBU. The conformally-weighted ψ_{ABCD} corresponds to the gravitational/curvature content carried across the boundary by V1 retarded propagation; the Killing-spinor κ^{AB} corresponds to the locally-symmetric subspace of V5's cross-channel coupling near the boundary; the closure $\nabla^{AA'}\phi_{AB} = 0$ corresponds to the substrate-side V5 cross-chain memory closure. The numerical coefficient $4\pi G$ is INHERITED from Meissner–Penrose 2025's Schwarzschild calibration. The substrate-graph derivation of a substrate-side conserved integral structurally equivalent to the twistor theorem is **OPEN** (audit row 13).

3.5 V5 cross-boundary memory signatures (localized + extended)

Per Meissner–Penrose 2025 §IV (eqs. 56–59), $\delta T/T \sim 3M_{gc}/(4\pi(\eta^c_{LS})^3 c^3 a^3_{LS}\,\rho_{LS}) \sim 10^{-3}$ from cross-boundary mass injection of $\sim 10^{15} M_\odot$. The observational anchor is the Hawking-spot temperature elevation at CMB-spot centers (An–Meissner–Nurowski–Penrose 2020; Planck data).

Localized identification (Hawking spots): under SCBU + V5 cross-boundary content, pre-boundary cluster-scale substrate events transfer event-density via V5 cross-chain memory across the boundary, sourcing localized post-boundary participation-density perturbations $\delta\rho^C$. Under Paper_003 substrate-frequency Born identification, $\delta\rho^C$ translates to CMB temperature elevation $\delta T/T$ at the corresponding spatial location. Numerical 10^{-3} INHERITED; mechanism IDENTIFIED.

Extended identification (Big Ring, Grand Arc): Big Ring (radius ∼200 Mpc, Lopez–Clowes–Williger 2024) and Grand Arc (∼1 Gpc, Lopez–Clowes–Williger 2022) are interpretable as cross-boundary V5 correlation imprints from pre-boundary substrate events at extended spatial scale (cluster-scale collisions in the diluting pre-boundary substrate per Meissner–Penrose 2025 §IX, eqs. 108–112). Same substrate mechanism as Hawking spots, scaled to extended spatial extent. Substrate-graph derivation of the extended-scale spatial scales from substrate parameters is **OPEN** (audit row 14).

3.6 Conformal matching ↔ scale-free participation geometry

CCC's conformal rescaling at $\mathscr{X}$ (Meissner–Penrose 2025 eqs. 6, 28) is the GR-side statement that absolute scale information is structurally absent at the boundary. In ED, this is the SC-4.x scale-collapse content (§3.1): no remaining structural content at the boundary distinguishes scales. Same boundary phenomenon, different vocabulary; CCC supplies explicit conformal-rescaling machinery, ED supplies the substrate-side justification (event-density dilution → SC-4.x scale-collapse → no remaining local content to encode absolute scale).

3.6.1 Uniqueness of the contrast-zero state — why the crossover gluing is forced, not postulated

CCC postulates that the conformally-rescaled future infinity of one aeon *matches* the conformally-rescaled Big Bang of the next (Meissner–Penrose 2025 §II): both surfaces are conformally invariant, so the continuation across $\mathscr{X}$ is consistent. Consistency, however, is weaker than identity. "Both are conformally invariant" does not by itself establish that the two boundary states are the *same* state — and CCC takes the match as a postulate (the conformal factor is glued by hand at $\mathscr{X}$).

Under the ED contrast ontology (Facts paper, "When Contrast Becomes Fact": a fact is a committed contrast; determinateness is commitment), the match is not postulated but forced. At extreme dilution the participation-density and committed-contrast content vanish (§3.1); the boundary state is the contrast-zero state. By the relational ontology, the contrast-zero state is **unique**: with no committed contrast there are no relations, and with no relations there remain no parameters in which two such states could differ (an isolated relatum carries no determination — "one in isolation is none"). There is therefore exactly *one* contrast-zero state. The end of an aeon and the beginning of the next are not two conformally-similar surfaces that must be glued by fiat; they are the single contrast-zero state, entered from above (dilution, §3.1) and exited from below (P12 threshold re-entry, §3.7). CCC's matching *assumption* becomes, under the contrast ontology, an *identity*: the gluing is a consequence, not an input.

This is the substrate-side justification for the one structural element CCC must assume — that the two ends coincide. It does not derive CCC's conformal-rescaling machinery; it grounds the *uniqueness* that machinery relies on. Tier: form-identification conditional on the contrast ontology (A→position / D-via-I; audit row 18), not a substrate-graph derivation.

3.7 Next-aeon ignition ↔ P12 threshold re-entry

Above the boundary, substrate event-density crosses Θ_{ED} (P12) from below; V1 + V5 + Paper_072 individuation reactivate; the substrate is in a globally-homogeneous, kernel-active, individuated initial state — the substrate-side counterpart of the next aeon's Big Bang. Whether this re-entry follows from the substrate-graph dynamics (committing ED to cyclic continuation) or is a separate substrate event (consistent with single-aeon termination) is **OPEN** (audit row 17).

3.8 Remnant relics and the contrast-zero boundary — a forced result

ED's black-hole arc *forces* stable Planck-mass remnants as the Hawking-evaporation endpoint: under V5 UV-cutoff dynamics the Hawking temperature reaches $\omega_c = c/\ell_P$ at $M \to M_\star = c_\star \ell_P$ and the evaporation rate vanishes — Scenario A (complete evaporation) is dynamically excluded, Scenario C (stable remnant) forced (Paper_041 + Paper_048 + Paper_040). Remnants are a within-aeon relic-matter component of the cosmic energy budget (Paper_049). This raises a sharp consistency question for the contrast-zero boundary (§3.1, §3.6.1): a surviving relic is a localized committed-contrast structure — a residual mass, hence a residual scale — and a relic population at $\mathscr{X}$ would obstruct *both* ED's contrast-zero closure *and* CCC's conformal closure (Meissner–Penrose 2025 requires no residual rest-mass at conformal infinity; the eventual disappearance of rest-mass is among CCC's most-scrutinized assumptions).

Two observations bound the issue.

First, the relics do not source the Hawking-point energy. The $\delta T/T \sim 10^{-3}$ signal (§3.5) is the *integrated* evaporation of a supermassive black hole ($\sim 10^{15} M_\odot$), released over the whole evaporation history and carried across $\mathscr{X}$ as common-cause continuation (audit row 22); the remnant is the final Planck-mass residue, energetically negligible against that signal. The remnant question is therefore independent of §3.5 and concerns only residual contrast at the boundary.

Second, ED has not committed to *eternal* remnants. Paper_041 makes remnant stability explicitly *conditional on V5-cutoff persistence* and declines to claim cosmological-timescale stability. The boundary is precisely the regime in which the V5 cross-chain kernel collapses ($b \to 0$; §3.1, §3.2): the cutoff that stabilizes the remnant is a *cross-chain* finite-memory structure, and a maximally isolated relic in a substrate whose cross-chain participation density has gone to zero has nothing left to sustain it.

This release is forced, not optional — the order-of-limits is not symmetric. The Scenario-C freeze is *not* an intrinsic property of the remnant. The UV cutoff $\omega_c = c/\ell_P$ that halts the final evaporation is *supported by V5 cross-chain correlation structure* (Paper_040, P-V5-UV-Cutoff), and V5 is **essentially a two-chain operator**: its correlation requires a *pair* of distinct participating chains (Paper_090 §3.2, "the two-chain character is essential"). The near-horizon cutoff is specifically the V5 handshake between the remnant's chain and the ambient near-horizon substrate vacuum structure, at the substrate-scale memory time $\tau_{V5}^{\mathrm{BH}} = \ell_P/c$ (Paper_090 §5.2). At a Planck-mass remnant ($M_\star \sim \ell_P$, of order one substrate cell) there is no *internal* partner chain; the stabilizing handshake is necessarily with the ambient. The contrast-zero boundary is defined by the vanishing of exactly that ambient ($\rho_C \to 0$; §3.1): in the limit there is no partner chain within the substrate-scale V5 correlation length $\ell_{V5} \sim \ell_{\mathrm{ED}}$ of the remnant, so the cutoff has no support; by Paper_041's own condition ("stability conditional on V5-cutoff persistence") the freeze lifts and the remnant completes its evaporation. P11 is preserved (the committed content is radiated, directionally, not un-committed); the realized record is carried forward as common-cause (A1). This also reconciles with ED-10 permanence: relic self-isolation is an *intra-aeon* mechanism (isolation from a populated exterior); at $\mathcal{X}$ the exterior itself vanishes, so there is nothing to self-isolate *from.* Dissolution is forced because the mechanism that would sustain the relic — a substrate-scale two-chain V5 cutoff — is precisely the structure the boundary's local emptying destroys. **Row 21 resolves toward dissolution (D-via-I).**

This is the substrate-side content of the assumption CCC must *make* — where the last rest-mass goes at conformal infinity. In ED it is not an assumption but a consequence: the evaporation-halt that produced the relics was a two-body handshake with a substrate the conformal boundary dissolves. The remaining question is whether the dynamics can *evade* dissolution by an emergent route. Two are available, and both close from the V5 kernel's own form (Paper_090 §3.1–§3.2).

Remnant–remnant pairing — surviving relics V5-stabilizing one another rather than the generic ambient — is closed by geometry. The V5 envelope $F_{V5}(\sigma/\ell_{V5}^2, \cdot)$ is bounded and decaying outside $\ell_{V5} \sim \ell_{\mathrm{ED}}$ (Paper_090 §3.1, §3.4), the substrate scale. Relics descend one per supermassive black hole, are already separated by cosmological distances, and the late-aeon de Sitter expansion drives every pair separation $\sigma \to \infty \gg \ell_{V5}$, so the substrate-scale pairing correlation $\to 0$. The long-range correlations at $\xi_{canonical}$ that *do* survive (they carry the Hawking-point common-cause signal, §3.3) are a different τ_{V5} regime and do not reproduce the substrate-scale cutoff $\omega_c = c/\ell_P$ — they cannot re-freeze the relic. No mutual handshake survives.

Internal restructuring — a relic hosting its own V5 partner — is closed by the substrate floor. V5 correlates two chains at *distinct loci* (Paper_090 §3.1); $M_\star = c_\star \ell_P$ with $c_\star \sim O(1)$ is of order one substrate cell, and P08 makes ℓ_{ED} the minimum edge length. A single-cell relic has no two distinct internal loci to host the handshake.

A third point must be explicit, because irreversibility makes it non-trivial. The ambient plays *two* roles for the relic: it is the V5 partner that *supplies* the cutoff (whose removal lifts the freeze, above) and it is the *destination* for the radiated content (P11 forbids erasing committed content; the V5 leak transfers it, it is never un-committed). Removing the ambient removes both — inviting the objection that, with no destination, the relic cannot radiate and so stays frozen. It does not: the content radiates into the diluting substrate at the crossover and is carried forward as common-cause (A1; row 22), so the relic's final release is part of the cross-boundary fossil, not a loss. Dissolution and the common-cause signal are the same process, and irreversibility is respected — content transferred, never erased.

With both emergent routes closed and the destination accounted for, row 21 is settled at the structural tier (the tier of the rest of the paper), not merely argued. The single residual is the generic limit of any analytic argument: an *unforeseen* emergent route. A direct substrate-graph simulation would surface one if it existed — but the program's certified discrete engine enforces monotone-irreversible commitment (the Bits substrate, B7) with no release channel, so testing relic *release* would require V5-leak machinery that engine does not implement. The faithful closure is therefore the analytic one above, not a simulation on existing tooling.

4 Audit Table

#	Step	Label	Notes
1	P02, P03, P04, P07, P09, P11, P12, P13	P	Paper_087.
2	Upstream inheritance block	I	Paper_028, Paper_072, Paper_089, Paper_090, Paper_092, Paper_093, Papers SC-4.x, Meissner–Penrose 2025, Lopez–Clowes–Williger.
3	P-CCC-Identification (naming convention)	P	Definitional per Paper_095 §2.3.
4	Substrate homogeneity in extreme-dilution limit	D-via-I	P02+P04+Papers SC-4.x scale-collapse; §3.1.
5	$\mathscr{X} \leftrightarrow$ SCBU identification	A→position	Substrate-graph derivation OPEN (row 14).
6	Horizon-reset = P12 ED-threshold transition	D-via-I	P11 + P12 + Paper_028; §3.2.
7	GWE $\leftrightarrow$ V1/V5 kernel-dominated regime	D-via-I	Paper_089 + Paper_090 + Meissner–Penrose 2025 §V; §3.3.
8	$\eta^G \leftrightarrow \xi_{canonical}$ identification	A→position	Quantitative consolidation pending Route A.
9	Twistor mass-cons. $\leftrightarrow$ V5 cross-boundary memory	A→position	Substrate-graph analog OPEN (row 14).
10	Hawking-spot $\delta T/T \sim 10^{-3} \leftrightarrow$ V5 signature	D-via-I	Paper_090 + Meissner–Penrose 2025 §IV + Paper_003; §3.5.
11	Big Ring / Grand Arc $\leftrightarrow$ extended V5 signature	D-via-I	Paper_090 + Meissner–Penrose 2025 §IX + Lopez–Clowes–Williger.
12	Conformal matching $\leftrightarrow$ SC-4.x scale-collapse	D-via-I	Papers SC-4.x + Meissner–Penrose 2025; §3.6.
13	Next-aeon ignition state structure	D-via-I	P12 + Paper_072 + Paper_089 + Paper_090 reactivation.
14	Substrate-graph derivation of V5 conservation analog	OPEN	§3.4.
15	Substrate-graph derivation of extended-scale transfer	OPEN	§3.5.
16	$\xi_{canonical}$ value from substrate (Route A)	OPEN	Standing OPEN across SC arc.
17	Cyclic continuation vs single-aeon termination	OPEN	§3.7.
18	Uniqueness of the contrast-zero state (crossover gluing forced, not postulated)	A→position	§3.6.1; conditional on the contrast ontology (Facts paper).
19	Remnant relics as within-aeon Scenario-C endpoint ($M_\star = c_\star \ell_P$)	I	§3.8; from the BH arc (Paper_041 / Paper_048 / Paper_040).
20	Remnant release at $\mathscr{X}$ via V5/b-collapse; contrast-zero closure preserved (P11 dilution-not-erasure; A1 common-cause)	A→position	§3.8; contingent on row 21.
21	Order-of-limits at $\mathscr{X}$ resolved toward **dissolution**: Scenario-C cutoff is a two-chain V5 handshake (Paper_090 §3.2/§5.2); local emptying removes the partner, the freeze lifts, the minimal relic evaporates	D-via-I	§3.8; Paper_090 + Paper_040 + Paper_041 + §3.1. Pairing + internal-restructuring routes closed analytically; residual = unforeseen route.

#	Step	Label	Notes
22	Cross-boundary signatures = common-cause continuation, not controlled transmission	D-via-I	§3.4 / §3.5 + A1 capacity-zero.
23	Verdict M3 (form-IDENTIFIED + value-INHERITED)	A→position	Per Paper_095 §3.3.

Zero pure-D rows. Audit content matches **M3 (form-IDENTIFIED + value-INHERITED)**: all substrate-side substantive rows are composition (D-via-I) or structural identification (A→position) of upstream content; no new substrate-graph derivation. Four named OPEN items (rows 14, 15, 16, 17); former row 21 (relic order-of-limits) resolved toward dissolution at D-via-I, with both emergent escape routes closed analytically (§3.8).

5 Falsification Criteria

- **F1 (sharpest):** Observed Hawking-spot magnitude or angular distribution at preferred CMB locations inconsistent with $\delta T/T \sim 10^{-3}$ under the V5 cross-boundary identification — beyond V5 finite-memory parameter freedom. Refutes §3.5 localized identification.
- **F2:** Persistence of curvature-scale information across the substrate dilution limit inconsistent with SC-4.x scale-collapse — substrate or observational regime in which absolute scale remains structurally encoded after dilution. Refutes §3.1 + §3.6.
- **F3:** Non-decay of V5 cross-chain correlations at $\xi_{canonical}$ in the late-aeon dilution limit. Once Route A closes, the quantitative consolidation $\eta^G \leftrightarrow \xi_{canonical}$ becomes itself a sharp test. Refutes §3.3.
- **F4:** Big Ring / Grand Arc structural scales admitting only standard within-aeon explanation under any V5 finite-memory parameter consistent with the rest of the corpus. Refutes §3.5 extended identification.
- **F5:** Demonstration that the cross-boundary signatures (Hawking points, twistor mass-conservation content) carry *controllable / structured* information across $\mathcal{X}$ beyond what shared-substrate common-cause continuation permits — i.e. a genuine signaling channel across the boundary. Refutes the A1 common-cause reclassification (§3.4 / §3.5, audit row 22) and the determinability-boundary identification (§3.2).
- **F6:** Evidence that the Scenario-C remnant cutoff is sustained *without* an ambient partner chain — a single isolated remnant self-supporting the V5 UV cutoff (contradicting Paper_090 §3.2's essential two-chain character). This blocks the §3.8 dissolution mechanism, leaves a residual relic-contrast population at $\mathcal{X}$, and refutes the contrast-zero closure for the ED-CCC reading (the ED-non-cyclic reading, row 17, may survive with a relic background).

6 Position Statement

This paper sits within the **substrate-ontology research-program lineage** ('t Hooft cellular-automaton; Wolfram Ruliad; causal-set program), **not** within the operational-reconstruction lineage (Hardy 2001, CDP 2011, Coecke-Kissinger). Methodology per Paper_095 (form-FORCED / value-INHERITED).

The paper supplies the substrate-side identification of CCC's continuum content (Meissner–Penrose 2025) with substrate content already present in the ED corpus. The identification does not derive CCC from substrate primitives, does not reduce ED to CCC, and does not commit

ED to cyclic cosmology: ED's SCBU is consistent with both ED-CCC (cyclic-continuation) and ED-non-cyclic readings, with audit row 17 OPEN. The genuine substrate content is the structural correspondence under P-CCC-Identification (rows 4–13); numerical content ($\delta T/T \sim 10^{-3}$, η^G, twistor coefficient $4\pi G$, Hawking-spot diameters, Big-Ring/Grand-Arc scales, $\xi_{canonical}$ value) is INHERITED.

Two §-level additions sharpen the form identification without moving the verdict. §3.6.1 grounds CCC's crossover-gluing postulate in the **uniqueness** of the contrast-zero state: the two aeon-ends are not two conformally-similar surfaces matched by fiat but *one state*, an identity under the contrast ontology. §3.8 confronts ED's forced Scenario-C remnants (Paper_041) with the contrast-zero boundary and **resolves** the relic order-of-limits toward dissolution: because the Scenario-C cutoff is an essentially two-chain V5 handshake with the ambient (Paper_090 §3.2/§5.2), the boundary's local emptying removes the partner and the freeze lifts — so the disappearance of residual rest-mass that CCC *assumes* is, in ED, *derived* (D-via-I, row 21; both emergent escape routes closed analytically in §3.8). This is the paper's one place where ED does not merely *identify* CCC content but *supplies* it.

Verdict: M3 (form-IDENTIFIED + value-INHERITED) per Paper_095. Four named OPEN items (rows 14–17); the relic order-of-limits (former row 21) is resolved toward dissolution (D-via-I), with both emergent escape routes closed analytically (§3.8). Closing row 16 (Route A) consolidates rows 8, 11 quantitatively; closing rows 14, 15 supplies the substrate-graph counterparts of MP25's twistor theorem and the extended-scale correlation transfer; closing row 17 commits ED to cyclic-continuation or single-aeon-termination cosmology. The relationship between ED and CCC is otherwise not competition between alternative reconstructions — it is structural correspondence at shared observational anchors.